

Integral Cancellation for Distributional Divergences on the Plane

Abstract

We prove that an integrable distributional divergence of a square-integrable vector field on the plane has zero integral. The proof constructs smooth expanding cutoffs, uses the scale invariance of their Dirichlet integral in two dimensions, and applies an integrable tail criterion. No estimate of the physical pressure or global-existence assertion for a compressible flow is included.

1 The cancellation criterion

All integrals are taken with respect to Lebesgue measure on $\mathbb{R}^2$, and $|x|$ denotes the Euclidean norm. The cancellation argument in Fan–Lü–Wang [1, Lemma 3.3] motivates the following elementary analytic criterion.

Lemma 1.1 *Let $f, h \in L^1(\mathbb{R}^2)$ be real-valued, with $h \geq 0$ almost everywhere. Let $\chi_n : \mathbb{R}^2 \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, be measurable. Suppose that constants $B, C \geq 0$ satisfy*

$$|\chi_n(x)| \leq B \quad (n \in \mathbb{N}, x \in \mathbb{R}^2), \quad \chi_n(x) \longrightarrow 1 \quad \text{for a.e. } x. \quad (1.1)$$

If, for every $n \in \mathbb{N}$,

$$\left| \int_{\mathbb{R}^2} f \chi_n \, dx \right| \leq C \left(\int_{\{|x| \geq n\}} h \, dx \right)^{1/2}, \quad (1.2)$$

then $\int_{\mathbb{R}^2} f \, dx = 0$.

Proof Put $E_n = \{x \in \mathbb{R}^2 : |x| \geq n\}$ and $H_n = \int_{E_n} h \, dx$. Each E_n is measurable. Since $h \in L^1(\mathbb{R}^2)$, its restriction to E_n is integrable, and $H_n \geq 0$. For each fixed x , the indicator $\mathbf{1}_{E_n}(x)$ is eventually zero. Hence

$$h \mathbf{1}_{E_n} \longrightarrow 0, \quad |h \mathbf{1}_{E_n}| \leq |h| \in L^1(\mathbb{R}^2).$$

The dominated convergence theorem gives

$$H_n = \int_{\mathbb{R}^2} h \mathbf{1}_{E_n} \, dx \longrightarrow 0. \quad (1.3)$$

On the other hand, each $f \chi_n$ is measurable up to a null set and

$$|f \chi_n| \leq B |f| \in L^1(\mathbb{R}^2).$$

Thus all testing integrals in (1.2) are integrals of integrable functions. By (1.1), $f \chi_n \rightarrow f$ almost everywhere. A second application of dominated convergence yields

$$I_n := \int_{\mathbb{R}^2} f \chi_n \, dx \longrightarrow I := \int_{\mathbb{R}^2} f \, dx. \quad (1.4)$$

Combining (1.2) and (1.3), we obtain

$$0 \leq |I_n| \leq C \sqrt{H_n} \longrightarrow 0.$$

Equation (1.4) also implies $|I_n| \rightarrow |I|$. Uniqueness of the limit gives $|I| = 0$, as asserted. $\square$

Remark 1.1 The estimate (1.2) is a substantive hypothesis. An application to a momentum equation must establish it from admissible testing, integrability, and control of the flux at infinity. Lemma 1.1 does not assert that these requirements hold for a particular compressible solution, nor does it establish a global existence theorem.

2 Distributional divergence in two dimensions

Theorem 2.1 *Let $f \in L^1(\mathbb{R}^2; \mathbb{R})$ and $V \in L^2(\mathbb{R}^2; \mathbb{R}^2)$. Suppose that*

$$\int f \varphi \, dx = - \int V \cdot \nabla \varphi \, dx \quad \text{for every } \varphi \in C_c^\infty(\mathbb{R}^2). \quad (2.1)$$

Then $\int f \, dx = 0$.

Proof Choose $\eta \in C_c^\infty(\mathbb{R}^2)$ with $0 \leq \eta \leq 1$, $\eta = 1$ on B_1 , and $\text{supp } \eta \subset \bar{B}_2$. For $n \geq 0$ put $r_n = n + 1$ and $\chi_n(x) = \eta(x/r_n)$. These are admissible tests in (2.1), and $\chi_n \rightarrow 1$ pointwise. The chain rule gives $\nabla \chi_n(x) = r_n^{-1} \nabla \eta(x/r_n)$, so the change of variables $x = r_n y$ yields

$$\int |\nabla \chi_n|^2 \, dx = r_n^{-2} r_n^2 \int |\nabla \eta(y)|^2 \, dy =: A^2 < \infty. \quad (2.2)$$

Moreover, $\nabla \chi_n = 0$ for $|x| < r_n$. The product $f \chi_n$ is integrable since $|\chi_n| \leq 1$, and $V \cdot \nabla \chi_n$ is integrable by Hölder's inequality. Testing and using (2.2), we obtain

$$\begin{aligned} \left| \int f \chi_n \, dx \right| &\leq \left(\int_{|x| \geq r_n} |V|^2 \, dx \right)^{1/2} \left(\int |\nabla \chi_n|^2 \, dx \right)^{1/2} \\ &\leq A \left(\int_{|x| \geq n} |V|^2 \, dx \right)^{1/2}. \end{aligned} \quad (2.3)$$

Apply Lemma 1.1 with $h = |V|^2$, $B = 1$ and $C = A$ to obtain the conclusion. $\square$

Remark 2.1 The cancellation depends on the planar scaling in (2.2). For a compressible momentum equation, identifying the tensor rows as L^2 fluxes and the material force as an L^1 function is a separate analytic application. The theorem does not bound either quantity uniformly in time.

References

1. X. Fan, B. Lü, X. Wang, Hardy Space and Exponential Decay of Inhomogeneous Navier–Stokes System in $\mathbb{R}^2$, arXiv:2610.04884v1 (2026).